

**Telegram @satuzbhunters**

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A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The area of the base of each of the two pyramids is  $100\text{ cm}^2$ . Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in  $\text{cm}^2$ ?

- To study fluctuations in the leaf water potential, samples of wood were taken from 26 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 2.00 centimeters. If this cube has a mass of 5.12 grams, what is the density of this cube in grams per cubic centimeter?

The figure shown consists of a rectangle and a semicircle, where the length of  $\overline{WX}$  is 20 units and the length of  $\overline{WZ}$  is 5 units. The diameter of the semicircle is  $\overline{WX}$ . Which of the following expressions represents the area, in square units, of the figure?

- A)  $20 + 5 + 10\pi$       C)  $20 \times 5 + 50\pi$   
B)  $20 \times 5 + 10\pi r$       D)  $20 \times 5 + 100\pi$

4.

Square  $P$  has a side length of  $x$  inches. Square  $Q$  has a perimeter that is 32 inches greater than the perimeter of square  $P$ . The function  $f$  gives the area of square  $Q$ , in square inches. Which of the following defines  $f$ ?

- A)  $f(x) = (x + 8)^2$
- B)  $f(x) = (x + 32)^2$
- C)  $f(x) = (32x + 8)^2$
- D)  $f(x) = (32x + 32)^2$

5.

|                           | Volume<br>(cubic units) |
|---------------------------|-------------------------|
| Right circular cylinder A | $112\pi$                |
| Right circular cylinder B | $3,024\pi$              |

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 4 units. The surface area of right circular cylinder A is  $k\pi$  square units, and the surface area of right circular cylinder B is  $n\pi$  square units, where  $k$  and  $n$  are constants. What is the value of  $n - k$ ? (The surface area of a right circular cylinder with radius  $r$  and height  $h$  is  $2\pi r^2 + 2\pi rh$ .)

6.

To estimate the proportion of a population that has a certain characteristic, a random sample was selected from the population. Based on the sample, it is estimated that the proportion of the population that has the characteristic is 0.2, with an associated margin of error of 0.08. Based on this estimate and margin of error, which of the following is the most appropriate conclusion about the proportion of the population that has the characteristic?

- A) It is plausible that the proportion is between 0.12 and 0.28.
- B) It is plausible that the proportion is less than 0.12.
- C) The proportion is exactly 0.2.
- D) It is plausible that the proportion is greater than 0.28.

7.

A researcher is designing a study to investigate the average number of hours a student at a high school spends reading per day. The researcher will report the average number of hours a student at the high school spends reading with an associated margin of error. The researcher is considering using a random sample of either 105 or 210 students from the high school. Which of the following could be the most likely effect of using the larger random sample compared to the smaller random sample?

- A) The reported margin of error would be lower.
- B) The reported margin of error would be higher.
- C) The reported average number of hours would be lower.
- D) The reported average number of hours would be higher.

8.

A proposal for a construction project was included on a city election ballot. A TV news report stated that 8 times as many people voted in favor of the proposal as people who voted against it. An internet article reported that 24,500 more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

9.

In a random sample of 276 residents in a certain building, 9 residents reported that they take the train to work. If there are a total of 1,380 residents in this building, which of the following is the best estimate of the number of building residents who take the train to work?

- |       |          |
|-------|----------|
| A) 9  | C) 276   |
| B) 45 | D) 1,095 |

10.

The values in data sets  $X$  and  $Y$  are shown in the table.

|              |    |    |    |    |    |    |    |    |    |
|--------------|----|----|----|----|----|----|----|----|----|
| Data set $X$ | 13 | 13 | 14 | 14 | 15 | 16 | 16 | 17 | 17 |
| Data set $Y$ | 2  | 2  | 3  | 3  | 4  | 5  | 5  | 6  | 6  |

The standard deviation of data set  $X$  is  $g$ , and the standard deviation of data set  $Y$  is  $s$ . Which of the following statements about the standard deviation of the data sets is true?

- A)  $g < s$
- B)  $g > s$
- C)  $g = s$
- D) The relationship between  $g$  and  $s$  cannot be determined.

11.

| Value | Data set $A$ frequency | Data set $B$ frequency |
|-------|------------------------|------------------------|
| 40    | 2                      | 8                      |
| 43    | 4                      | 7                      |
| 46    | 5                      | 5                      |
| 49    | 7                      | 4                      |
| 52    | 8                      | 2                      |

Data set  $A$  and data set  $B$  each consist of 26 values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

- A) The mean of data set  $A$  is greater than the mean of data set  $B$ .
- B) The mean of data set  $A$  is less than the mean of data set  $B$ .
- C) The mean of data set  $A$  is equal to the mean of data set  $B$ .
- D) There is not enough information to compare the means of the data sets.

12.

*Dataset A* : 2, 2, 2, 5, 5, 8, 11, 11, 14, 14, 14

*Dataset B* : 2, 5, 5, 8, 8, 8, 8, 11, 11, 14

Data set  $A$  and data set  $B$  each have 11 values. Which of the following statements best compares the median of data set  $A$  and the median of data set  $B$ ?

- A) The median of data set  $A$  is greater than the median of data set  $B$ .
- B) The median of data set  $A$  is less than the median of data set  $B$ .
- C) The medians of data sets  $A$  and  $B$  are equal
- D) There is not enough information to compare the medians.

13.

Data set  $F$  consists of 55 integers, where the value of each integer is between 150 and 260. Data set  $G$  consists of the same 55 integers in  $F$  as well as the integers 10. Which of the following must be less for data set  $G$  than for data set  $F$ ?

- I. The mean
  - II. The median
- A) I only  
B) II only  
C) I and II  
D) Neither I nor II

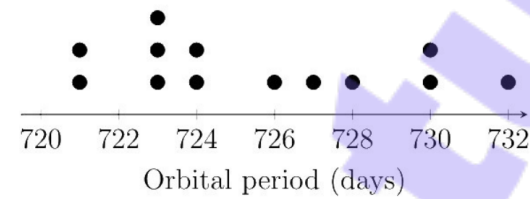
14.

| Number of queen ants | Number of colonies |
|----------------------|--------------------|
| 1                    | 11                 |
| 2                    | 9                  |
| 3                    | 7                  |
| 4                    | 6                  |
| 5                    | 6                  |

Scientists collected acorns that each housed a colony of a particular species and then analyzed each colony's structure. For these colonies, the table summarizes the number of queen ants per colony. For these colonies, what is the median number of queen ants per colony?

- A 6                      C 3  
B 4                      D 2

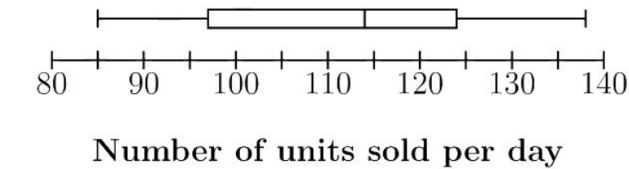
15.



A data set of the orbital periods, rounded to the nearest whole number of Earth days for 13 of Jupiter's moons is represented in the dot plot. An additional moon with an orbital period of 628 days is added to the original data set to create a new data set of 14 orbital periods. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

- A) The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is equal to the median of the original data set.
- B) The mean of the new data set is equal to the mean of original data set, and the median of the new data set is less than the median of the original data set.
- C) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.
- D) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

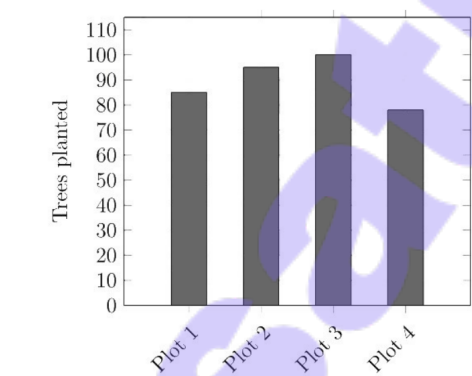
16.



A store owner recorded the number of units of a certain product that were sold per day for 28 days. What is the minimum number of units of this product sold per day for these 28 days?

17.

The bar graph shows the number of trees planted for 4 separate plots of land on a tree farm.



What is the maximum number of trees planted on one of these 4 plots of land?

- A) 100
- B) 95
- C) 85
- D) 78

18.

A researcher surveyed undergraduate students, graduate students, and postgraduate students. The number of undergraduate students surveyed was 7450% of the number of postdoctoral students surveyed and the number of graduate students surveyed was 24% of the undergraduate students surveyed. If there were 3576 graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?

19.

The positive number  $a$  is 2,352% of the sum of the positive numbers  $b$  and  $c$ , and  $b$  is 84% of  $c$ . What percent of  $b$  is  $a$ ?

- A) 5,152%                      C) 51.52%  
B) 2,436%                      D) 24.36%

20.

The mass of object A is 498% of the mass of object B, and the mass of object A is 0.830% of the mass of object C. If the mass of object C is  $p\%$  of the mass of object B, what is the value of  $p$ ?

21.

For the positive quantities  $h$ ,  $j$ , and  $k$ . 94% of  $h$  is equivalent to 47% of  $j$ , and  $j$  is equivalent to 30% of  $k$ . What percentage of  $k$  is  $h$ ? (Disregard the % sign when entering your answer. For example, if you answer is 39%, enter 39)

22.

A square map has a side length of 35 inches, and 1 inch on the map represents an actual distance of 17 miles. A smaller version of the same map is printed as a square with the side length 55% shorter than the side length of the previous map. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

- |          |          |
|----------|----------|
| A) 7.65  | C) 19.25 |
| B) 10.97 | D) 37.78 |

# SAT UZBEKISTAN

## Math Module-2

Time: 35 minutes

1.

Triangles  $ABC$  and  $DEF$  are congruent, where  $A$  corresponds to  $D$ , and  $B$  and  $E$  are right angles. The measure of angle  $A$  is  $71^\circ$ . What is the measure, in degrees, of angle  $F$ ?

2.

At a convention center, there are a total of 375 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.48, and the probability of selecting a visitor who is located in room B is 0.32. How many visitors are located in room C?

(A) 20



(B) 58



(C) 60



(D) 75



3.

$$g(m) = -0.04m + 16.6$$

The given function  $g$  models the number of gallons of gasoline that remains from a full gas tank in a car after driving  $m$  miles. According to the model, about how many gallons of gasoline are used to drive each mile?

(A) 0.04



(B) 16.6



(C) 25



(D) 415.0



4.

| $x$ | $h(x)$ |
|-----|--------|
| 0   | 1.46   |
| 2   | 2.32   |
| 4   | 3.68   |

The table shows the exponential relationship between the number of years,  $x$ , since Isela started training in pole vault, and the estimated height  $h(x)$ , in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where  $x \leq 4$ ?

(A)  $h(x) = 1.26(0.46)^x$  ☐

(B)  $h(x) = 1.26(1.46)^x$  ☐

(C)  $h(x) = 1.46(0.26)^x$  ☐

(D)  $h(x) = 1.46(1.26)^x$  ☐

5.

$$f(x) = 2,000(0.85)^x$$

A conservation scientist implemented a program to reduce a certain invasive insect population in an area. The given function estimates this insect species' population  $x$  years after 2008, where  $x \leq 8$ . Which of the following is the best interpretation of 2,000 in this context?

(A) The estimated percent decrease in the insect population for this species and area every 8 years after 2008 ☐

(B) The estimated percent decrease in the insect population for this species and area each year after 2008 ☐

(C) The estimated insect population for this species and area 8 years after 2008 ☐

(D) The estimated initial insect population for this species and area in 2008 ☐

6.

The cost to rent a bus from Company X is \$950 for the first 2 hours and an additional \$50 per hour for each hour after the first 2 hours. If the total cost to rent the bus from Company X for  $t$  hours, where  $t > 2$ , is \$1,250, which equation represents this situation?

(A)  $950(t - 2) + 50t = 1,250$  ☐

(B)  $950(2t) + 50t = 1,250$  ☐

(C)  $950 + 50(t - 2) = 1,250$  ☐

(D)  $950 + 50(2t) = 1,250$  ☐

7.

$$y = -\frac{1}{4}x^2 + x + 29$$

The given equation models a company's active projects over 6 months, where  $y$  is the estimated number of active projects  $x$  months after the end of April 2013, where  $0 \leq x \leq 6$ . Which statement is the best interpretation of the  $y$ -intercept of the graph of this equation in the  $xy$ -plane?

(A) At the end of April 2013, the estimated number of active projects was 0.

Ⓐ

(B) At the end of April 2013, the estimated number of active projects was 29.

Ⓑ

(C) At the end of May 2013, the estimated number of active projects was 0.

Ⓒ

(D) At the end of May 2013, the estimated number of active projects was 29.

Ⓓ

8.

$$17.5x + 28.75y = 530$$

Odalys ordered mulch and river rock, which cost a total of \$530, for her home. The given equation represents the relationship between the number of cubic yards of mulch,  $x$ , and the number of tons of river rock,  $y$ , Odalys ordered. How much more, in dollars, did a ton of river rock cost Odalys than a cubic yard of mulch?

9.

A hemisphere is half of a sphere. If a hemisphere has a radius of 20 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

(A) 800

Ⓐ

(B) 3,400

Ⓑ

(C) 12,600

Ⓒ

(D) 16,800

Ⓓ

10.

In  $\triangle ABC$ ,  $\angle B$  is a right angle and the length of  $\overline{BC}$  is 204 millimeters. If  $\cos A = \frac{3}{5}$ , what is the length, in millimeters, of  $\overline{AB}$ ?

Ⓐ 255

Ⓐ

Ⓑ 204

Ⓑ

Ⓒ 153

Ⓒ

Ⓓ 51

Ⓓ

11.

In a certain state, the population of pheasants, a type of bird, is estimated each year by counting the number of pheasants observed along certain roads in the state. On average, each year from 2005 to 2015 the number of pheasants counted per mile of road decreased by 3.5% of the number of pheasants per mile of road the previous year. Based on this average, if there were 6.93 pheasants per mile of road in this state in 2005, which of the following best approximates the number of pheasants per mile of road in 2015?

Ⓐ  $6.93(0.965)^{10}$ 

Ⓐ

Ⓑ  $6.93(0.035)^{10}$ 

Ⓑ

Ⓒ  $1.035(6.93)^{10}$ 

Ⓒ

Ⓓ  $0.965(6.93)^{10}$ 

Ⓓ

12.

To study fluctuations in leaf water potential, samples of wood were taken from 24 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 2.00 centimeters. If this cube has a mass of 3.04 grams, what is the density of this cube, in grams per cubic centimeter?

13.

| $x$ | $y$ |
|-----|-----|
| -11 | -33 |
| 6   | 52  |

The table shows two values of  $x$  and their corresponding values of  $y$ . The graph of the linear equation representing this relationship passes through the point  $(\frac{1}{4}, a)$ . What is the value of  $a$ ?

14.

Which system of linear equations has no solution?

(A)  $-5x + 9y = -8$   
 $5x - 9y = 8$

Ⓐ

(B)  $5x - 9y = 8$   
 $6x + 10y = 9$

Ⓑ

(C)  $5x - 9y = 8$   
 $-15x + 27y = -24$

Ⓒ

(D)  $-5x + 9y = 8$   
 $10x - 18y = 16$

Ⓓ

15.

In triangle  $ABC$ , the measure of angle  $A$  is  $48^\circ$ , the measure of angle  $B$  is  $90^\circ$ , and the measure of angle  $C$  is  $\left(\frac{k}{2}\right)^\circ$ . What is the value of  $k$ ?

(A) 42

Ⓐ

(B) 45

Ⓑ

(C) 84

Ⓒ

(D) 96

Ⓓ

16.

| Number of cars | Maximum number of passengers and crew |
|----------------|---------------------------------------|
| 4              | 164                                   |
| 6              | 242                                   |
| 8              | 320                                   |

The table shows the linear relationship between the number of cars,  $c$ , on a commuter train and the maximum number of passengers and crew,  $p$ , that the train can carry. Which equation represents the linear relationship between  $c$  and  $p$ ?

(A)  $39c - p = -8$

Ⓐ

(B)  $39c - p = 8$

Ⓑ

(C)  $39p - c = -8$

Ⓒ

(D)  $39p - c = 8$

Ⓓ

17.

$$g(x) = 3(16x - 17)$$

What is the  $y$ -coordinate of the  $y$ -intercept of the graph of  $y = g(x) - 5$  in the  $xy$ -plane?

☐ (A) -56



☐ (B) -51



☐ (C) 22



☐ (D) -17



18.

At the start of an experiment, the pressure inside a container was 19 pounds per square inch (psi). The pressure inside the container doubled every 28 minutes after the start of the experiment. Which equation represents this situation, where  $y$  is the pressure, in psi, inside the container  $x$  minutes after the start of the experiment?

☐ (A)  $y = 19(2)^{\frac{x}{28}}$



☐ (B)  $y = 19(28)^{2x}$



☐ (C)  $y = 19(x + 28)^2$



☐ (D)  $y = 19x^2 + 28$



19.

Quadrilateral  $P'Q'R'S'$  is similar to quadrilateral  $PQRS$ , where  $P$ ,  $Q$ ,  $R$ , and  $S$  correspond to  $P'$ ,  $Q'$ ,  $R'$ , and  $S'$ , respectively. The measure of angle  $P$  is  $45^\circ$ , the measure of angle  $Q$  is  $70^\circ$ , and the measure of angle  $R$  is  $90^\circ$ . The length of each side of  $P'Q'R'S'$  is 3 times the length of each corresponding side of  $PQRS$ . What is the measure of angle  $P'$ ?

☐ (A)  $15^\circ$



☐ (B)  $45^\circ$



☐ (C)  $60^\circ$



☐ (D)  $135^\circ$

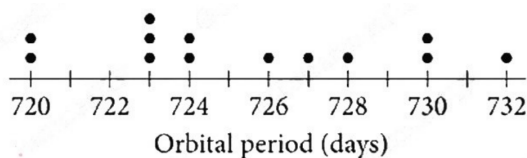


20.

$$x^2 + 11x + 19 = 0$$

One solution to the given equation can be written as  $x = \frac{-11 - \sqrt{k}}{2}$ , where  $k$  is a constant. What is the value of  $k$ ?

21.



A data set of the orbital periods, rounded to the nearest whole number of Earth days, for 13 of Jupiter's moons is represented in the dot plot. An additional moon with an orbital period of 628 days is added to the original data set to create a new data set of 14 orbital periods. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

(A) The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

Ⓐ

(B) The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is less than the median of the original data set.

Ⓑ

(C) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.

Ⓒ

(D) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

Ⓓ

22.

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by  $x$ . The product of 18 and the fourth odd integer is at most 22 less than the sum of the first and third odd integers. Which inequality represents this situation?

(A)  $18(x + 6) \leq x + (x + 4) - 22$

Ⓐ

(B)  $18(x + 6) \geq 22 - (x + (x + 4))$

Ⓑ

(C)  $18(x + 4) \leq x + (x + 3) - 22$

Ⓒ

(D)  $18(x + 4) \geq 22 - (x + (x + 3))$

Ⓓ